

Section 3.3

Exercise (3.3.1). Let F be a finite field of order q and let $n \in \mathbb{Z}^+$. Prove that $|GL_n(F) : SL_n(F)| = q - 1$

Proof: Suppose F is a finite field of order q . Let $n \in \mathbb{Z}^+$ and consider $GL_n(F)$ and $SL_n(F)$.

Recall from exercise 3.1.35 of Homework 3 where we showed that $GL_n(F) \trianglelefteq SL_n(F)$ showing that the map $f : GL_n(F) \rightarrow F^*$ defined by $f(A) = \det(A)$ was a homomorphism (albeit a dubious argument stating $\det(AB) = \det(A)\det(B)$ by some property of determinants). We continued by applying the First Isomorphism theorem to state that $GL_n(F)/S_n(F) \cong F^*$. Since F^* has order $q - 1$ this demonstrates that $|GL_n(F) : SL_n(F)| = q - 1$. $\square$

Exercise (3.3.3 - Stefano Fochesatto). Prove that if H is a normal subgroup of G of prime index p then for all $K \leq G$ either (i) $K \leq H$ or (ii) $G = HK$ and $|K : K \cap H| = p$.

Proof: Suppose H is a normal subgroup of G of prime index p and $K \leq G$. Note that since $H \trianglelefteq G$ we know that $N_G(H) = G$ and further that $K \leq N_G(H)$. By the Second Isomorphism Theorem it then follows that $KH/H \cong K/K \cap H$, and $H \leq KH \leq G$. We demonstrated in class that if $H \leq KH \leq G$ then $|G : H| = |G : KH||KH : H|$. Since $KH/H \cong K/K \cap H$ we know that $|KH : H| = |K : K \cap H|$. Therefore by substitution we get,

$$|G : KH||KH : H| = |G : KH||K : K \cap H| = |G : H| = p.$$

(i) Suppose that $|G : KH| = p$ and $|K : K \cap H| = 1$. Therefore for every $k \in K$, $kK \cap H = 1K \cap H$. So $k \in K \cap H$ and therefore $k \in H$. Thus $K \leq H$.

(ii) Suppose that $|G : KH| = 1$ and $|K : K \cap H| = p$. Therefore for every $g \in G$, $gKH = 1KH$. Therefore $g \in KH$ and $G \leq KH$. Since $KH \leq G$ from the Second Isomorphism Theorem, it follows that $G = KH$. Recall that Proposition 14 gives us that $HK = KH$ if either is a subgroup of G . Hence $G = HK$. $\square$

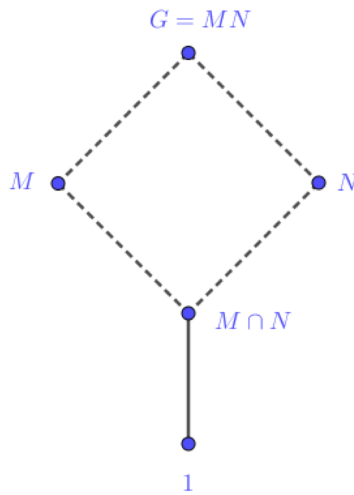
Exercise (3.3.7). Let M and N be normal subgroups of G such that $G = MN$. Prove that $G/(M \cap N) \cong (G/M) \times (G/N)$ [Draw the Lattice].

Proof: Suppose M and N are normal subgroups of G such that $G = MN$. Consider the mapping $\varphi : G \rightarrow (G/M) \times (G/N)$ defined by $\varphi(g) = (gM, gN)$. Note that $\varphi(ab) = (abM, abN) = (aM, aN)(bM, bN) = \varphi(a)\varphi(b)$ (the operation $aMbM = abM$ is well defined by Proposition 5, Section 3.1). Thus φ is a homomorphism and clearly any element $g \in M \cap N$ will map to (M, N) , the identity. Hence $\ker(\varphi) = M \cap N$. Since $G = MN$ every $g \in G$ can be written as a product $g = mn$ for some pair $m \in M$ and $n \in N$. Thus for any pair of cosets (g_1M, g_2N) we know there exists some $m_1, m_2 \in M$ and $n_1, n_2 \in N$ such that $(g_1M, g_2N) = (m_1n_1M, m_2n_2N)$. Since M is a normal subgroup we can simplify,

$$(g_1M, g_2N) = (m_1n_1M, m_2n_2N) = (Mm_1n_1, m_2n_2N) = (Mn_1, m_2N) = (n_1M, m_2N) = \varphi(n_1m_2).$$

Thus φ is a surjective homomorphism. Applying the First Isomorphism Theorem we get that $G/(M \cap N) \cong (G/M) \times (G/N)$.

FIGURE 1. Lattice Diagram for G . Dashed lines denote normal relationship.



□

Section 3.4

Exercise (3.4.4). Use Cauchy's Theorem and induction to show that a finite abelian group has a subgroup of order n for each positive divisor n of its order.

Proof: Let G be a finite abelian group. Clearly if $|G| = 1$ then the statement holds, there exists a subgroup, with order of each divisor of 1. Also note that for prime orders of G the statement holds, namely subgroups 1 and G , since p has no other divisors.

We continue via strong induction on $|G| = n$. Suppose that for all k from $1 \leq k < n$ there exists subgroups $H_i \leq G_k$ where $|H_i|$ is a divisor of $|G_k| = k$.

Let d be a divisor of G_n . If d is prime then by Cauchy's Theorem it follows that there exists some element $x \in G_n$ such that $|x| = d$, therefore $|\langle x \rangle| = d$ and we have our subgroup. If d is not prime then there exists some prime p such that $d = jp$ where $j \in \mathbb{Z}$. By Cauchy's Theorem there exists some $x \in G_n$ such that $|\langle x \rangle| = p$. Now consider $G_n/\langle x \rangle$ and since all our groups are finite recall that by Lagrange's Theorem, $|G_n/\langle x \rangle| = |G_n|/|\langle x \rangle| = n/p$. Note that $|G_n/\langle x \rangle| < |G_n| = n$ and has order with divisor j . Thus by the inductive hypothesis, there exists some subgroup $H \leq G_n/\langle x \rangle$ such that $|H| = j$. Since G_n is abelian $\langle x \rangle$ is normal in G_n , therefore by the Lattice Isomorphism Theorem, H is of the form $H/\langle x \rangle$ for some $H \leq G_n$. Note that by Lagrange's Theorem $|H/\langle x \rangle| = |H|/|\langle x \rangle| = j$, and therefore $|H| = jp = d$. $\square$

Exercise (3.4.5). Prove that subgroups and quotient groups of a solvable group are solvable.

Proof: Let G be a solvable group and $H \leq G$. Since G is solvable we know that there exists a chain of subgroups,

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \cdots \trianglelefteq G_s = G$$

such that G_{i+1}/G_i is abelian for $0 \leq i \leq s-1$. Consider the set of subgroups of H , $H_i = (H \cap G_i)$ with $0 \leq i \leq s$. For H to be solvable we must show that $H_i \trianglelefteq H_{i+1}$ and H_{i+1}/H_i is abelian. Let $h_1 \in H_i$ and $h_2 \in H_{i+1}$. Note that since $G_i \trianglelefteq G_{i+1}$

we know that $h_2h_1h_2^{-1} \in G_i$, and clearly $h_2h_1h_2^{-1} \in H$. Thus $h_2h_1h_2^{-1} \in H_i$ and we conclude that $H_i \trianglelefteq H_{i+1}$.

Next consider the quotient group $H_{i+1}/H_i = (H \cap G_{i+1})/(H \cap G_i)$. Note that since $H_i \trianglelefteq H_{i+1} \trianglelefteq H$ we know that $H_i = H \cap G_i = H_{i+1} \cap G_i$. By substitution we get $H_{i+1}/H_i = H_{i+1}/H_{i+1} \cap G_i$. Clearly $H_{i+1} \leq N_g(G_i) = G$ and thus by the Second Isomorphism Theorem we know that $H_{i+1}/H_{i+1} \cap G_i \cong H_{i+1}G_i/G_i$, $G_i \trianglelefteq H_{i+1}G_i$, and $H_{i+1}G_i \leq G$. Note that $H_{i+1}G_i/G_i = (G_{i+1} \cap H)G_i/G_i \leq G_{i+1}/G_i$ since $(G_{i+1} \cap H)G_i \leq G_{i+1}$. Since $H_{i+1}G_i/G_i$ is a subgroup of an abelian group it must also be abelian, and therefore H_{i+1}/H_i is abelian. Thus H is solvable

Let $N \trianglelefteq G$ consider $\bar{G} = G/N$. By The Fourth Isomorphism Theorem (We need $N \trianglelefteq G$ for this) there exists a bijection from the set of subgroups of G which contain N and the set of subgroups of G/N . Since G is solvable there must exist a chain of normal subgroups of $\bar{G}$ where $\bar{G}_i = G_i/N$ such that $N \leq G_i$,

$$1 = \bar{G}_0 \trianglelefteq \bar{G}_1 \trianglelefteq \bar{G}_2 \trianglelefteq \cdots \trianglelefteq \bar{G}_s = \bar{G}.$$

Now we must demonstrate that $\bar{G}_{i+1}/\bar{G}_i$. Note that $N, G_i \trianglelefteq G_{i+1}$ (This comes from $N \trianglelefteq G$ and $N \subseteq G_i$) therefore by the Third Isomorphism Theorem we conclude that $\bar{G}_{i+1}/\bar{G}_i = (G_{i+1}/N)/(G_i/N) \cong G_{i+1}/G_i$. Since G is solvable G_{i+1}/G_i is abelian and therefore $\bar{G}_{i+1}/\bar{G}_i$ is also abelian. Thus G/N is solvable. $\square$

Exercise (3.4.6). Prove (1) of the *Jordan – Hölder Theorem* by induction on $|G|$. If G is a finite group with $|G| \neq 1$ then G has a composition series.

Proof: Let G be a finite group with $|G| \neq 1$. We will proceed by induction on $|G| = n$.

Consider the base case, $|G| = 2$. Clearly the group G is composed of two elements $G = \{1, g\}$ with $g^{-1}g = 1$. Consider the composition series, $1 \leq G$.

Now suppose for all $|G| = k$ such that $2 \leq k < n$ there exists a composition series. Let $|G| = n$. Consider a proper normal subgroup H . By the Induction Hypothesis H

has a composition series,

$$1 \leq H_0 \leq H_1 \leq \cdots \leq H_m = H$$

Note that all $H_i \leq G$. Now consider $\bar{G} = G/H$. Since H is a proper normal subgroup we know that $|\bar{G}| < n$. Thus by the Induction Hypothesis there exists a composition series,

$$H \leq \bar{G}_1 \leq \bar{G}_2 \leq \cdots \leq \bar{G}_i = \bar{G}.$$

By the Fourth Isomorphism Theorem there exists a bijection φ from the set of subgroups of G which contain H and the set of subgroups of $\bar{G}$. So let $\varphi^{-1}(\bar{G}_i) = G_i$ and note it also follows from the Fourth Isomorphism Theorem since $\bar{G}_i \leq \bar{G}_{i+1}$ we know that $G_i \leq G_{i+1}$. Finally note that by the Third Isomorphism Theorem we know that $\bar{G}_{i+1}/\bar{G}_i \cong G_{i+1}/G_i$. Thus the composition factors for G_i are simple, giving us the following composition series,

$$H \leq G_1 \leq G_2 \leq \cdots \leq G_i = G.$$

Finally by substitution we get the following composition series for G

$$1 \leq H_0 \leq H_1 \leq \cdots \leq H_m = H \leq G_1 \leq G_2 \leq \cdots \leq G_i = G.$$

□

Section 3.5

Exercise (3.5.3). Prove that S_n is generated by $\{(i(i+1)) | 1 \leq i \leq n-1\}$

Proof: Let G be the group generated by $\{(i(i+1)) | 1 \leq i \leq n-1\}$. We will proceed by induction on $|S_n| = n$. Consider the base case, let $n = 2$. Clearly G contains all elements from S_2 i.e. (12) and $(12)^2$. Suppose that for all $2 \leq k < n$, $S_k = G_k = \langle \{(i(i+1)) | 1 \leq i \leq k-1\} \rangle$. Let $a \in S_n$ and note that a can be written as a product of transpositions. Let all transpositions that comprise a be of the

form $(a_1 a_2)$. If $a_1, a_2 \in [n-1]$ then clearly by the induction hypothesis $(a_1 a_2) \in G_{n-1} \leq G_n$. Similarly if $a_1 = n-1$ and $a_2 = n$ then by definition $(a_1, a_2) \in G_n$. Consider the case where $a_1 \in [n-1]$ and $a_2 = n$. With conjugates we see that $(a_1 n) = ((n-1)n)(a_1(n-1))((n-1)n) \in G_n$. Since every transposition of S_n is in G_n , it follows that $a \in G_n$. Thus S_n is generated by G_n . $\square$

Exercise (3.5.6). Show that $\langle (13), (1234) \rangle$ is a proper subgroup of S_4 . What is the isomorphism type of this subgroup?

Proof: Let $G = \langle (13), (1234) \rangle$. Note that $(13)^2 = (1234)^4 = 1$, and $(1234)(13) = (13)(1234)$.

Note that these generators function identically to $r, s \in D_{2(4)}$. Consider the map $\varphi : D_{2(4)} \rightarrow G$ defined by $\varphi(r) = (1234)$ and $\varphi(s) = (13)$. This map is clearly an isomorphism, hence $|G| = |D_{2(4)}| = 8$. With $|G| = 8$, G must be a proper subgroup of S_4 since $|S_4| = 24$. $\square$

Exercise (3.5.10). Find a composition series for A_4 . Deduce that A_4 is solvable.

Solution: Consider the subset of A_4 , $G = \{1, (12)(34), (13)(24), (14)(23)\}$. We will demonstrate that $G \trianglelefteq A_4$. First note that for any pair $(ab)(cd), (ad)(bc) \in G$ the product is the third $(ab)(cd)(ad)(bc) = (ac)(db)$. Since $(ac)(db) = (ca)(bd)$ we get that G is also abelian. Also note that each element is its own inverse. Therefore $G \leq A_4$. Note that for any $a \in A_4$ we can treat it as a bijective map from $a : [4] \rightarrow [4]$. Clearly any aGa^{-1} will result in permutation of the form $(ab)(cd)$ so $G \trianglelefteq A_4$. Since G is abelian we know all subgroups of G must be normal. Thus we get the following composition series (we can double check with the lattice diagram at the end of Section 3.5),

$$1 \leq \langle (12)(34) \rangle \leq G \leq A_4.$$

Note that $|\langle (12)(34) \rangle / 1| = 2$, $|G / \langle (12)(34) \rangle| = 2$, and $|A_4 / G| = |A_4| / |G| = 3$. Note that with the orders of every composition factor being even, they are all cyclic and therefore abelian. Thus A_4 is solvable. $\square$